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Quantitative Aptitude

Formula – Guide

Ratio & Proportion

Ratio and Proportion

Ratio

Ratio can be represented as, a to b or a : b. Here a and b could be any two quantities. Where 'a' is called the 'antecedent' and 'b' is called the 'consequent'.

Proportion

let us assume that, the two ratios are a : b & c : d are in proportion i.e. a : b :: c : d then,

Product of mean = Product of extremes

Thus, a : b :: c : d $\leftrightarrow$ (b × c) = (a × b)

Directly Proportion

If "a" and "b" are two quantities, then the direction proportion is written as $a \propto b$.

Inverse Proportion

The inverse proportion of two quantities, say "a" and "b" is represented by $a \propto (1/b)$.

Compound Ratio

Compounded ratio of the ratios , a : b and c : d can be written as, ac : bd.

Third Proportion

Third proportion of a and b = b^2/a

Mean Proportion

Mean proportion between a and b = $\sqrt{ab}$

Fourth Proportion

If a : b :: c : d or in other words a : b = c : d, then

Fourth proportion, d = $\frac{(b \times c)}{a}$

Properties of Ratio and Proportion

- **Invertendo:**

  Invertendo of a : b :: c : d is b : a :: d : c

  i.e. $\frac{a}{b} = \frac{c}{d}$, then $\frac{b}{a} = \frac{d}{c}$

- **Alternendo:**

  If a : b = c : d is a proportion then its alternendo is a : c = b : d; i.e. $\frac{a}{b} = \frac{c}{d}$, then $\frac{a}{c} = \frac{b}{d}$

- **Componendo:**

  If a : b = c : d is a proportion then its componendo is (a + b) : b :: (c + d) : d .

  i.e. $\frac{a}{b} = \frac{c}{d}$, then $\frac{(a+b)}{b} = \frac{(C+d)}{d}$

- **Dividendo:**

  If a : b = c : d is a proportion then its dividendo is (a - b) : b :: (c - d) : d .

  i.e. $\frac{a}{b} = \frac{c}{d}$, then $\frac{(a-b)}{b} = \frac{(C-d)}{d}$

- **Componendo and Dividendo:**

  If a : b = c : d is a proportion then its componendo and dividendo is

  (a + b) : (a – b) :: (c + d) : (c – d) .

  i.e. $\frac{a}{b} = \frac{c}{d}$, then $\frac{(a+b)}{(a-b)} = \frac{(C+d)}{(c-d)}$